

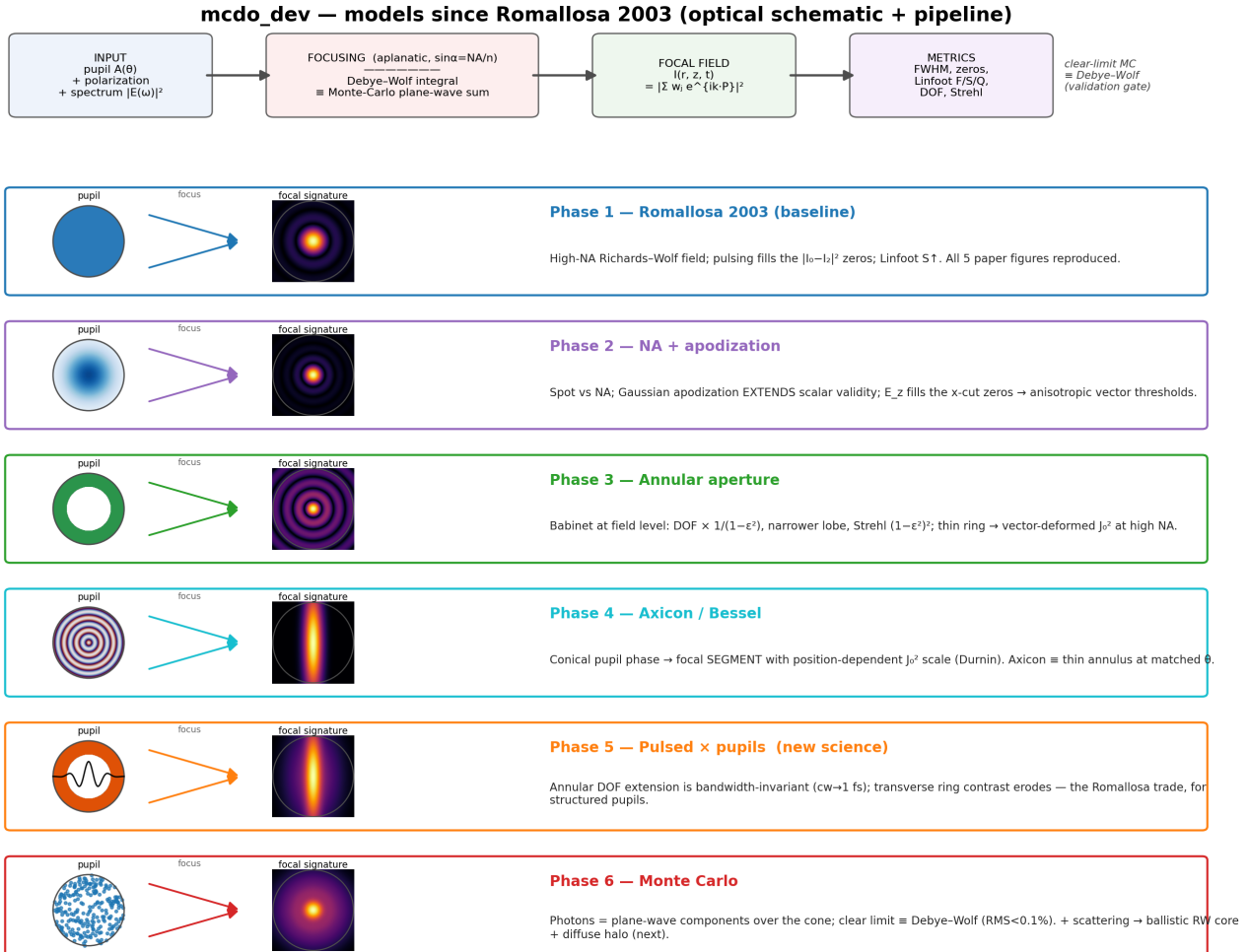
Review 06 — Monte-Carlo focal field (clear limit + initialization proof)

mcd_dev Phase 6a (output/06_monte_carlo/, docs/theory/06_monte_carlo.md)

2026-06-14

0. Where this sits — the pipeline

Every model since Romallosa runs the same pipeline: a pupil $A(\theta)$ (+ polarization, + spectrum) is focused by an aplanatic system — the Debye–Wolf integral *or* its Monte-Carlo plane-wave equivalent — into a focal field, then scored by FWHM / zeros / Linfoot / DOF / Strehl. The MC *clear limit equals* the Debye–Wolf integral; that equality is the validation gate this review establishes.

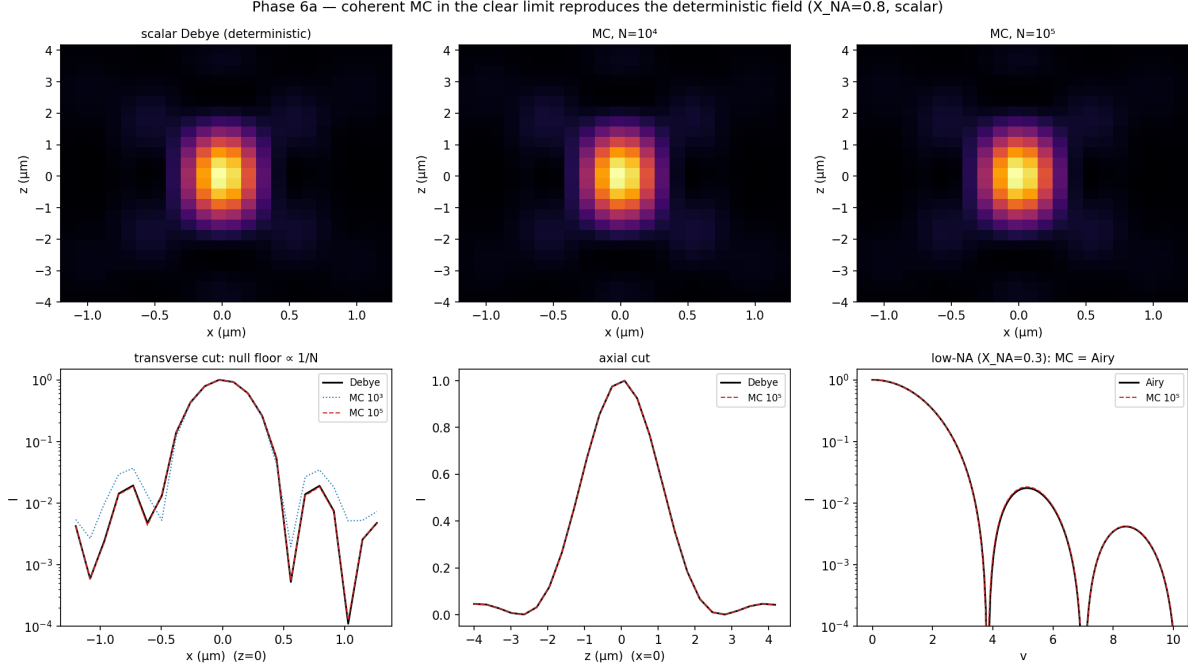


1. Why a photon is a plane wave, and the clear-limit gate

A photon is a plane-wave (angular-spectrum) component of the converging field, $U(P) = \sum_j w_j e^{ik_j \cdot P}$. With $\mu_s = 0$ this is a Monte-Carlo evaluation of the Debye–Wolf integral, so it must converge to Richards–Wolf ($X_{NA} = 0.8$, scalar):

N	RMS vs deterministic	null floor
10^3	0.0117	4.4×10^{-3}
10^4	0.0033	5.3×10^{-4}
10^5	0.0008	$\sim 5 \times 10^{-4}$

Null floor $\propto 1/N$ (random-walk residual at the true zeros); low-NA matches the analytic Airy to 0.0015.

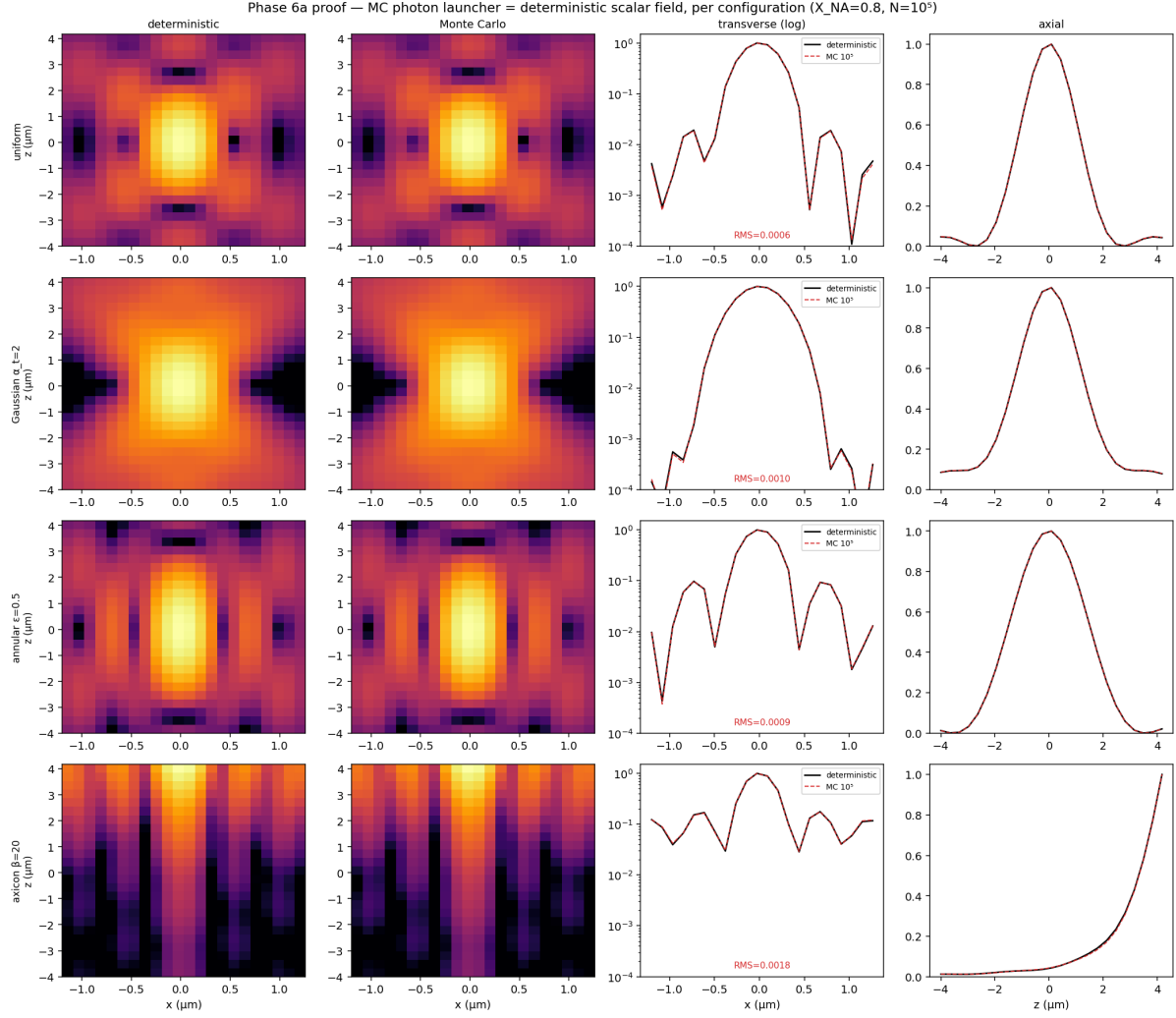


2. Per-configuration proof

Same scalar Debye integral, deterministic vs MC photons ($N = 10^5$, RMS over the x - z slice):

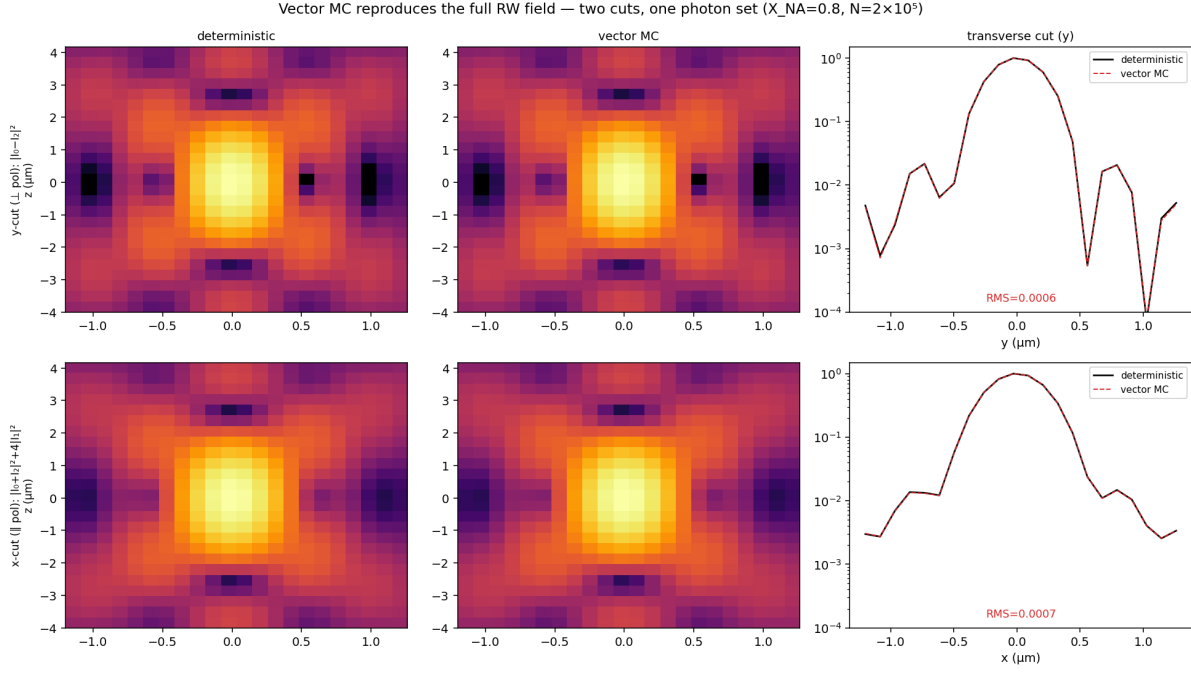
configuration	RMS
uniform	0.0006
Gaussian $\alpha_t = 2$	0.0010
annular $\varepsilon = 0.5$	0.0009
axicon $\beta = 20$	0.0018

All $< 0.2\%$: the launcher correctly encodes apodization (density), obstruction (support), and pupil phase (weight).



3. Vector field

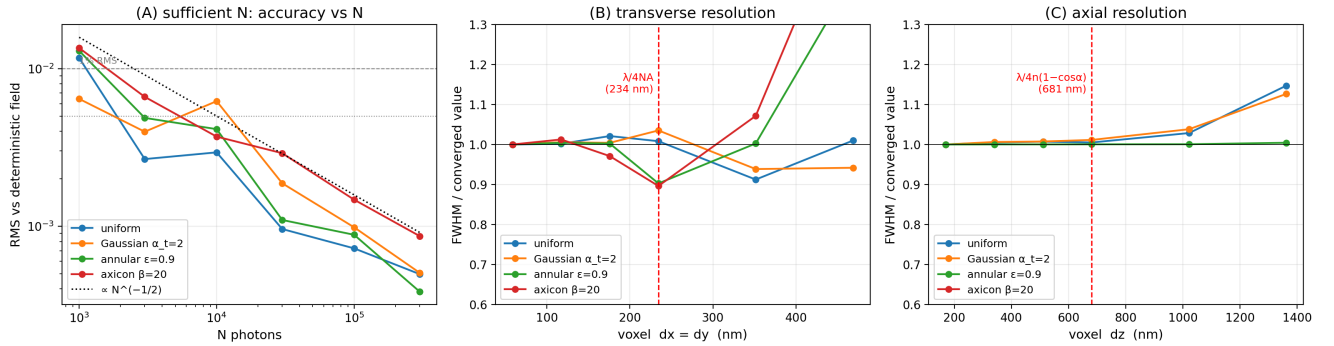
Photons carry the refracted 3-vector polarization (Novotny–Hecht Eq. 3.66); $|E_x|^2 + |E_y|^2 + |E_z|^2$ reproduces both Richards–Wolf cuts ($|I_0 - I_2|^2$ on the y -axis, $|I_0 + I_2|^2 + 4|I_1|^2$ on the x -axis), $\text{RMS} \sim 6 \times 10^{-4}$.



4. Sampling requirements — sufficient N and voxel size

Two *independent* requirements. **(A) Sufficient N** (statistical): $\text{RMS} \propto 1/\sqrt{N}$, null floor $\propto 1/N$; $N \approx 3 \times 10^3$ for 1% RMS, axicon needs $\approx 10^4$ for clean zeros. **(B) Voxel** (Nyquist on the *intensity*, twice the field frequency): $dx = dy \leq \lambda/(4\text{NA}) = 234 \text{ nm}$, $dz \leq \lambda/(4n(1 - \cos \alpha)) = 681 \text{ nm}$; we run at half (117 / 341 nm). FWHM recovery (normalised) is flat below the Nyquist voxel and degrades above it.

Phase 6a sampling requirements — sufficient N and voxel size (NA=0.8, $\lambda=750 \text{ nm}$, $n=1.3$)

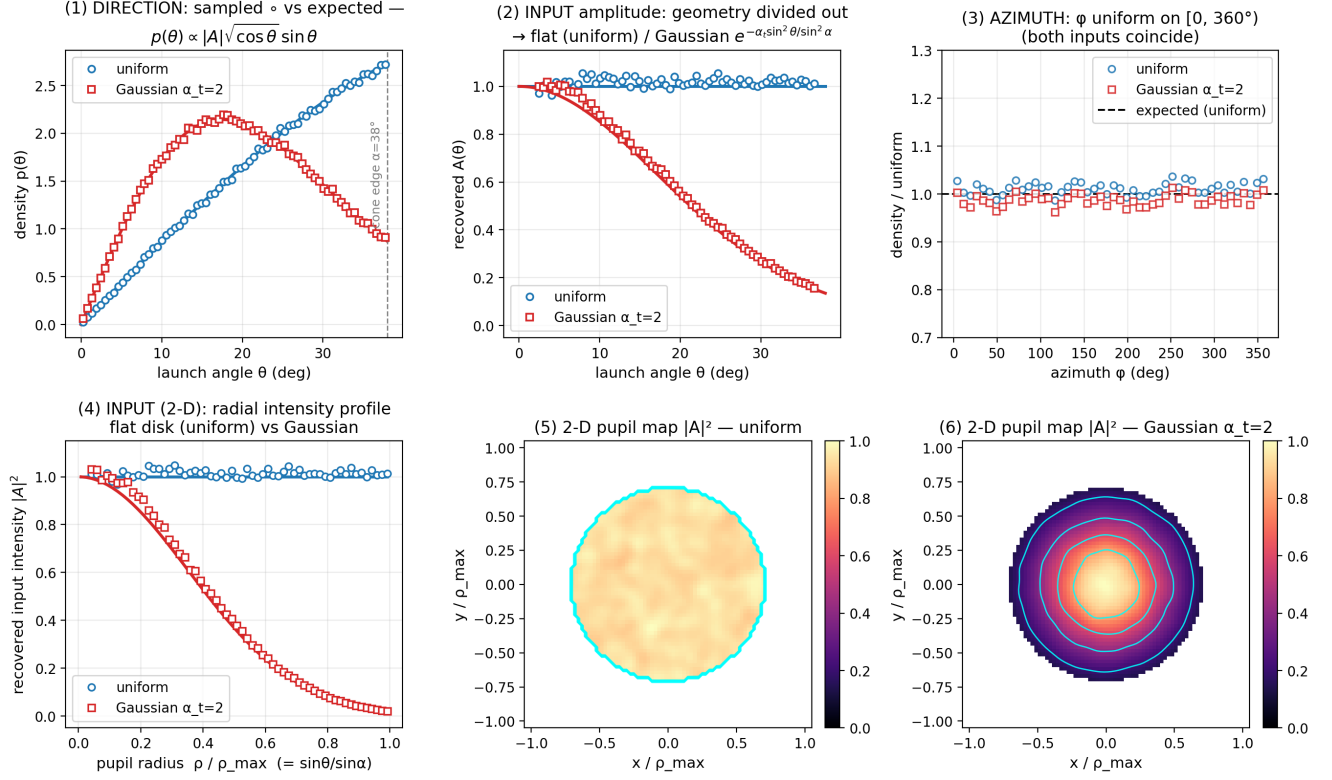


5. Launcher initialization proof (the pre-scattering prerequisite)

Why correct from first principles, not just distribution-matching: the plane-wave decomposition *is* Debye–Wolf theory; directions are confined to the cone $\theta \leq \alpha$ ($\sin \alpha = \text{NA}/n$) with statistical weight $A(\theta)\sqrt{\cos \theta}$ fixed by the input illumination $\times$ the aplanatic (Abbe sine) factor — energy conservation, not a fit. Location and direction are locked by $\rho = f \sin \theta$, and the importance-sampling estimator is provably unbiased, so $N \rightarrow \infty$ convergence is guaranteed *before* running. In the clear medium photons travel in straight lines (the wavefront normals).

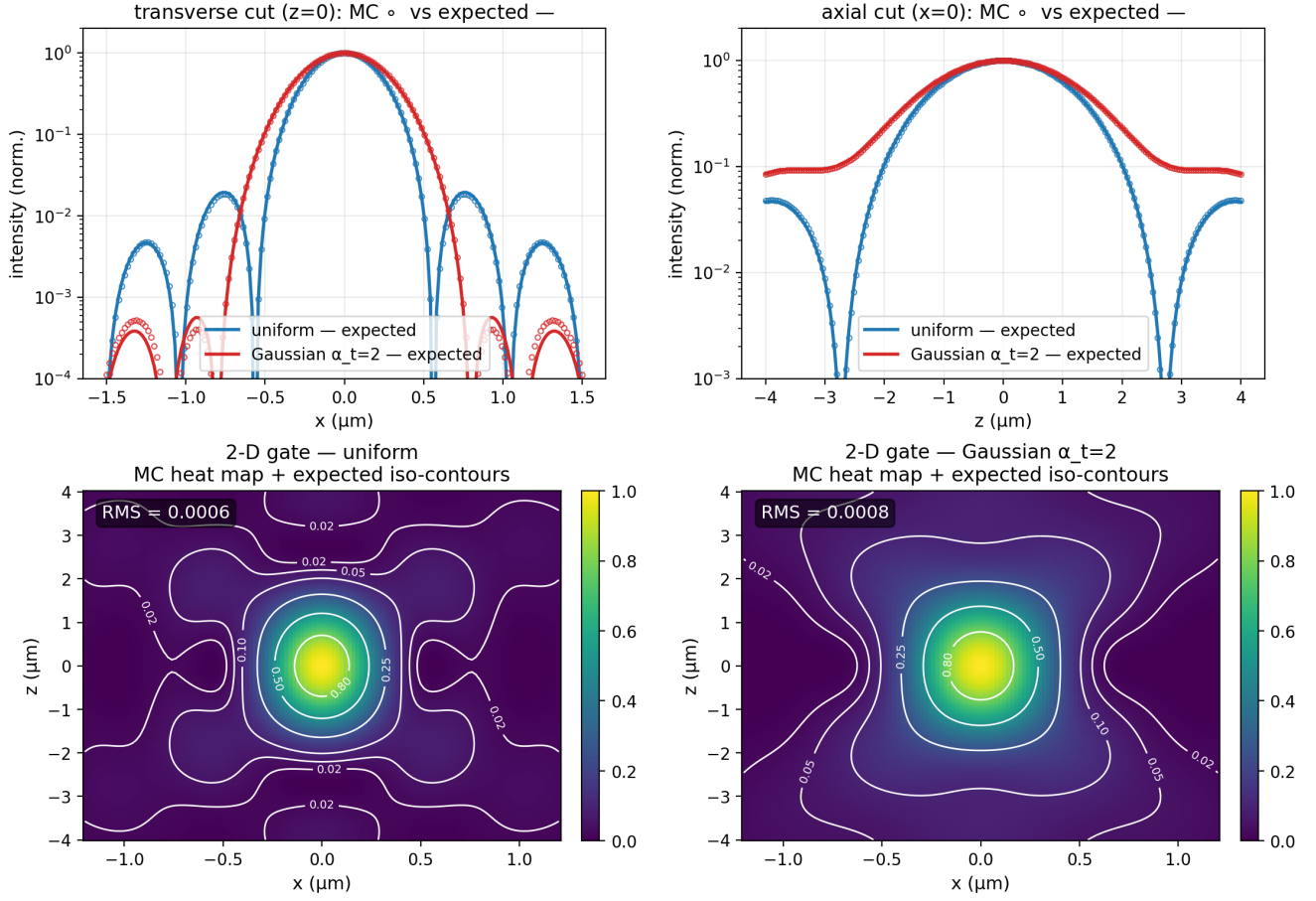
Distribution (uniform & Gaussian $\alpha_t = 2$): (1) sampled θ matches the Debye–Wolf density $|A|\sqrt{\cos \theta} \sin \theta$, KS = 0.0009 $\approx 1/\sqrt{N}$; (2) φ uniform (flatness 0.025; both inputs coincide); (3) recovered $A(\theta)$ flat / Gaussian; (4) recovered radial input intensity $|A|^2 = \text{flat disk} / \text{Gaussian spot}$.

Phase 6a — photon initialization is sampled correctly (clear limit, no scattering)



Intensity: MC matches the deterministic field on transverse & axial cuts (through peak, zeros, side-lobes) and on the x - z slice, RMS 0.0006–0.0008; the deterministic iso-value contours sit on the MC heat map.

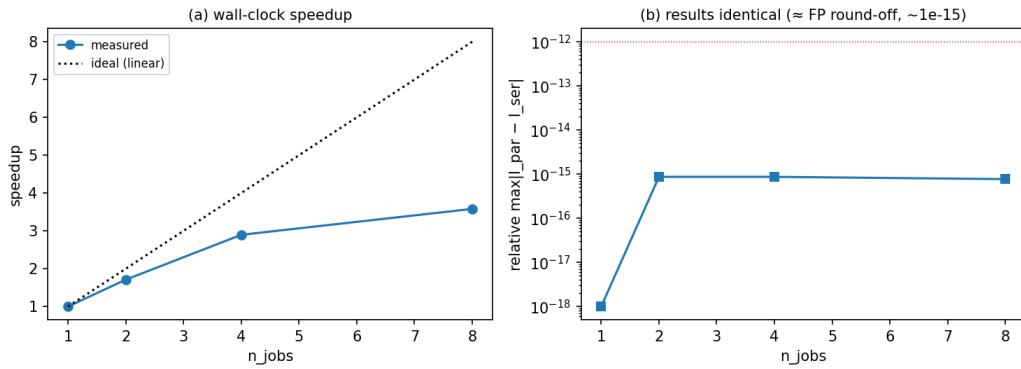
Phase 6a — intensity after MC launching matches the expected field (coherent clear limit)



6. Optional parallelism

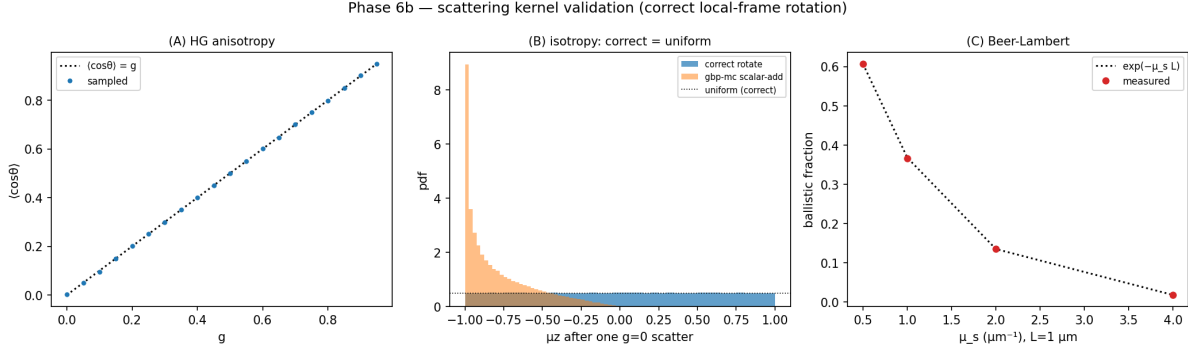
Partitioning photons across workers and summing the partial complex fields is identical to serial (the field is an additive sum): agreement to FP round-off (8.7×10^{-16} relative), speedup 1.7/2.9/3.6 $\times$ at 2/4/8 jobs.

MC parallelism — speedup and result identity ($N=2 \times 10^5$)



7. Scattering kernel — pre-validated; 6b next

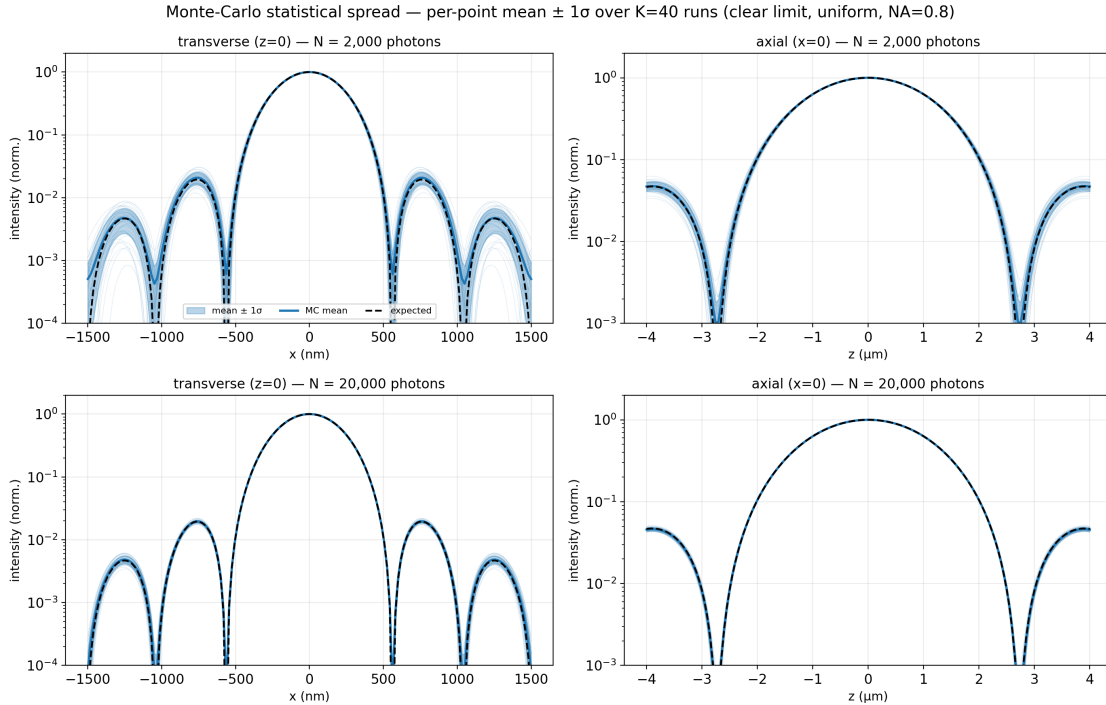
The kernel for the next phase is already checked in isolation: Henyey–Greenstein $\langle \cos \theta \rangle = g$; isotropic $\langle \mu_z^2 \rangle = 1/3$; unit-norm directions; Beer–Lambert ballistic fraction $\exp(-\mu_s L) = 0.607/0.367/0.135/0.018$ exact.



6b note. Because the launcher is certified, the only new physics is attenuation: the ballistic (coherent) field attenuates in *amplitude* as $\exp(-\mu_s L/2)$ (Beer–Lambert $\exp(-\mu_s L)$ in intensity), with the scattered energy forming a separate *incoherent* halo. $\mu_s \rightarrow 0$ recovers Phase 6a by construction.

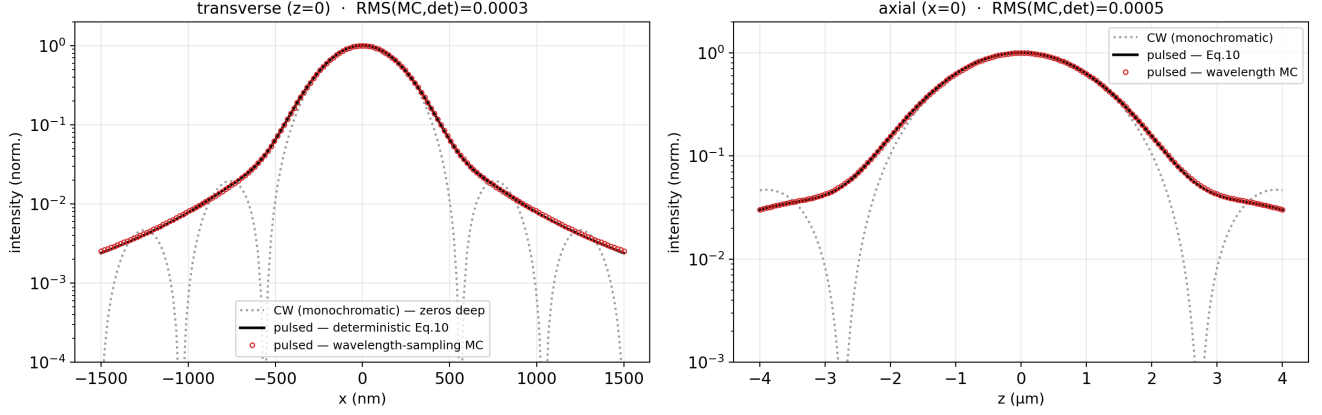
8. Addenda — statistical spread, wavelength MC, photon budgets

Statistical spread (`mc_variance.png`): per-point mean $\pm 1\sigma$ over $K=40$ runs (transverse & axial, $N=2k/20k$); the band is widest in the dark side-lobes/zeros and shrinks $\propto 1/\sqrt{N}$.



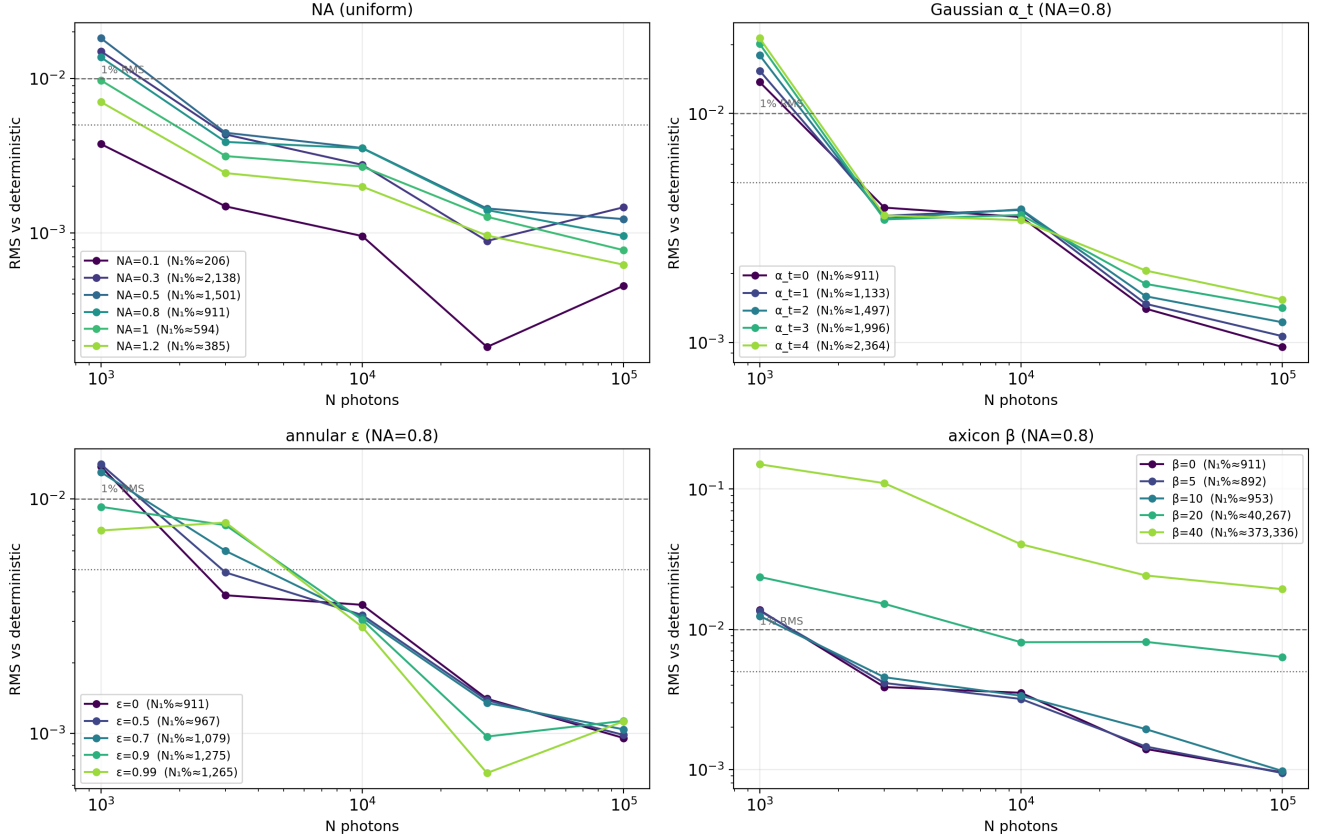
Per-photon wavelength MC (`verify_mc_pulsed.png`): drawing each photon's $\lambda \sim |E(\omega)|^2$ (coherent within a wavelength, incoherent across) reproduces the deterministic Eq. 10 to RMS $3\text{--}5 \times 10^{-4}$ and fills the CW zeros. Use it for pulsed light in *scattering* media (each λ sets its own μ_s); the deterministic sum is cheaper in a clear medium.

Per-photon wavelength-sampling MC vs deterministic Eq. 10 (clear limit, uniform, NA=0.8, $\tau=2$ fs)



Photons per model (`photons_per_model.png`): N for 1% RMS — NA 0.1–1.2 ~ 200 –2100; Gaussian α_t 0 $\rightarrow$ 4 $\sim 0.9k \rightarrow 2.4k$; annular ε 0 $\rightarrow$ 0.99 $\sim 0.9k \rightarrow 1.3k$; axicon $\beta=0/5/10 \sim 0.9k$ but $\beta=20 \approx 40k$ and $\beta=40 \approx 370k$. The cost is set by how fast the pupil *phase* winds (coherent cancellation), not by NA — and in the clear limit it is small because this is importance-sampled coherent quadrature of a smooth 2-D integral. Scattering (high-dimensional path space, an incoherent halo over many voxels, an exponentially small ballistic fraction at depth) is what pushes counts to the usual 10^6 – 10^8 .

Photons needed per model — RMS(MC – deterministic) vs N ($\lambda_c=750$ nm, $n=1.3$)

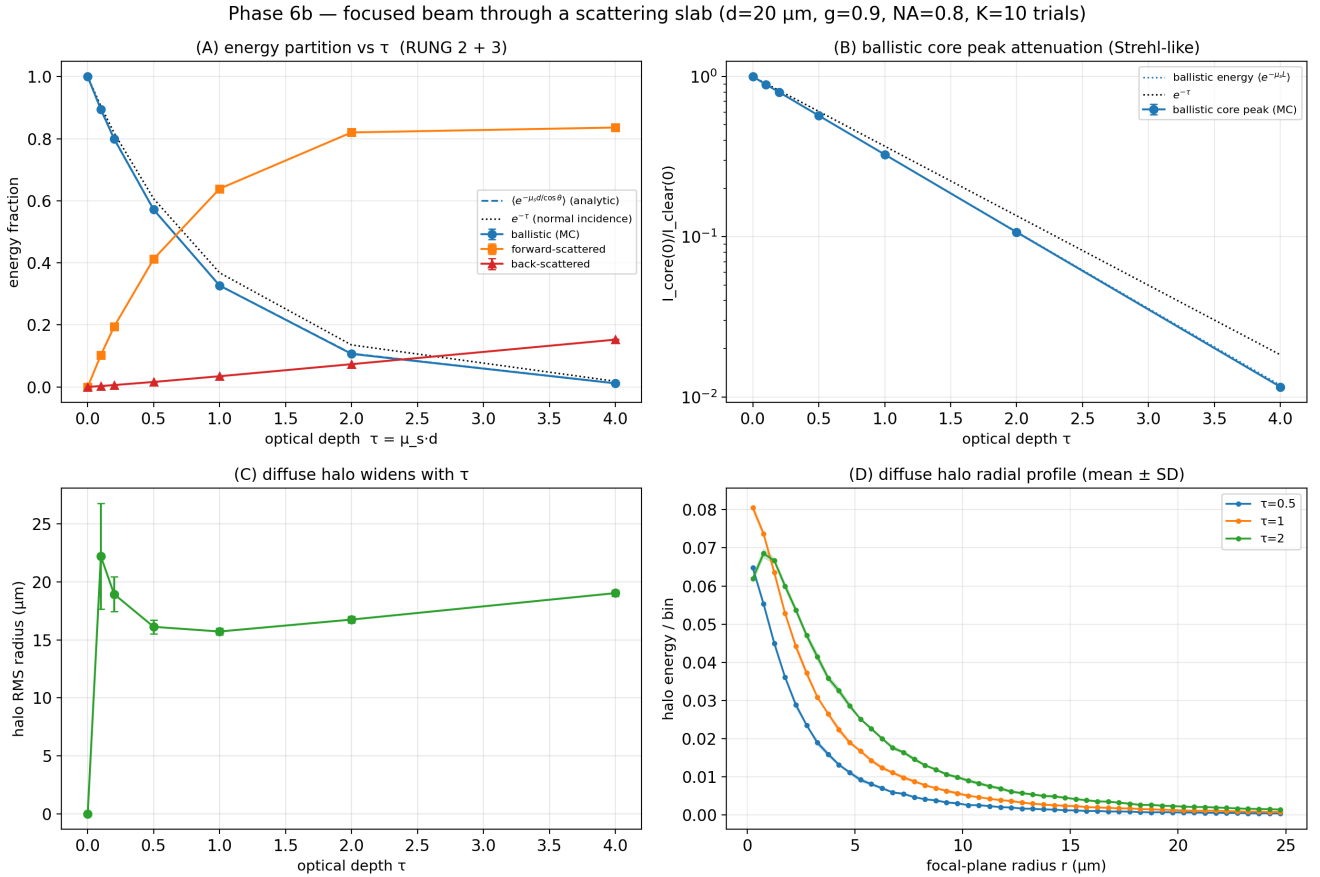


9. Phase 6b — focused beam through a scattering slab (first run, validated)

Slab $z \in [-d, 0]$ (focus at the back face), $d = 20 \mu\text{m}$, $g = 0.9$, $\text{NA}=0.8$, $K = 10$ trials. Each photon is the converging ray, entering at $(-d \tan \theta \cos \varphi, -d \tan \theta \sin \varphi, -d)$. The focal field splits into a **ballistic coherent core** (amplitude $e^{-\mu_s L/2}$, $L = d/\cos \theta$) and a **diffuse incoherent halo** (forward-scattered photons, Henyey–Greenstein walk). *All three validation runs pass:*

- **RUNG 1 continuity:** at $\mu_s = 0$ the core = the clear focus (peak ratio 1) and the halo = 0.
- **RUNG 2 Beer–Lambert:** the MC ballistic fraction equals the analytic cone average $\langle e^{-\mu_s d/\cos \theta} \rangle$ to < 0.001 across $\tau = \mu_s d \in [0, 4]$.
- **RUNG 3 energy:** ballistic + forward + back + absorbed = 1 to 2×10^{-16} .

Physics. As τ grows the ballistic focus dims ($E_{\text{ball}} = 0.89/0.57/0.33/0.011$ at $\tau = 0.1/0.5/1/4$, *below* the normal-incidence $e^{-\tau}$ because $d/\cos \theta > d$), forward-scatter dominates ($g = 0.9$; backscatter ≤ 0.15), the diffuse halo grows ($\sim 16\text{--}22 \mu\text{m}$ RMS), and the ballistic core *broadens* $\sim 5\%$ ($478 \rightarrow 501 \text{ nm}$) as steep rays are preferentially scattered. This is the corrected, error-barred analogue of Arjonillo Fig 4.1.



Next: validation runs 4–6 (single-scatter Born, diffusion asymptote, MCML cross-check); the axial halo; and vector + pulsed (λ -dependent μ_s) scattering.